

VALENTINO CARMONA

Backend Software Engineering | Computer Engineering Student

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PROFILE

Computer Engineering student (~75% completed, FIUBA/UBA) focused on Backend Software Engineering, with hands-on experience building and deploying backend systems in Java/Spring Boot, Node.js, and Python/Flask, and working with Databases such as PostgreSQL and MongoDB. I solved 87 algorithms and data structures problems on LeetCode (76.7th percentile). I have a solid software engineering foundation spanning requirements elicitation, architecture, design patterns, and testing, complemented by systems experience implementing the SDP/STUN/ICE protocols and concurrent RTP pipelines in Rust.

TECHNICAL SKILLS

Backend: Java Spring Boot · Node.js · Express · Flask · REST APIs · WebSockets · Socket.IO · JWT · HTTP/HTTPS

Databases: PostgreSQL · MongoDB · SQL · relational and NoSQL; familiarity with Neo4j and Cassandra

Languages: Java · Python · JavaScript · Rust; familiarity with Go · C/C++ · Smalltalk · Assembly

Systems & Networking: Linux/Unix · TCP/UDP · WebRTC · SDP · ICE · STUN · RTP

Software Engineering: OOP · SOLID · GoF design patterns · TDD · BDD · SDD · unit and integration testing · software architecture · requirements elicitation · C4 Model

Tools: Git · Docker · GitHub Actions · Postman · Swagger UI · JaCoCo · Cucumber · pytest · JUnit 5 · Jest · codecov

ACADEMIC PROJECTS

RoomRTC · Rust · WebRTC · SDP · STUN · ICE · RTP · Concurrency

FIUBA – Programming Workshop | 2025

- Implemented an SDP library from scratch in Rust, including *SessionDescription*, *MediaDescription*, *SessionOrigin*, *MediaType*, *Connection*, *Attribute* and *Codec*, with parsing, serialization, and unit tests.
- Contributed to implementing STUN (*RFC 5389*) and ICE candidate gathering through STUN requests using the standard library; proposed and implemented the STUN server configuration through the *.conf* file.
- Diagnosed and resolved a deadlock in the RTP pipeline caused by a global lock in the *session*, separated the *sender* and *receiver* locks, and proposed a multi-threaded architecture to the team with both components independent.

Balatro Clone · Java · SOLID · GoF · TDD/BDD · JUnit 5 · GitHub Actions 2025

FIUBA – Algorithms and Programming III

- Proposed and implemented a *JokerStrategy* hierarchy to encapsulate variable joker behaviors through *apply()*, with specialized strategies for effects on hands, score, discards, and multipliers, avoiding centralized conditionals and making the system easier to extend.
- Contributed to algorithmic poker-hand detection and the JUnit 5 test suite, configured CI with GitHub Actions, and applied SOLID principles in the team's design reviews over 4 functional iterations across 5 weeks.

PSA & PMTool · Spring Boot · PostgreSQL · REST APIs · C4 · Requirements Elicitation 2025–2026

FIUBA – Software Engineering I & Systems Development Management

- Contributed to requirements elicitation with stakeholders (CEO, Finance, Operations, Support) through 6 formal documented interview sessions, using Lean Inception and Personas analysis. Produced an SRS with a point-to-point bidirectional traceability matrix and a risk log.
- Designed the system architecture with Domain Storytelling (DDD) and the C4 Model (Context/Containers), and implemented the backend with Spring Boot and PostgreSQL, end-to-end from requirements to backend/API.

PERSONAL PROJECTS (all deployed on Render)

- Balatro Web Migration:** (Spring Boot · React/TS · JWT · Docker · JaCoCo 89%), REST API, Spring Security with HTTP headers, CORS, JWT, and OWASP Dependency-Check to detect known vulnerabilities in backend dependencies.
- API-CRUD:** REST API, 7 endpoints, JWT authentication (global middleware), and CRUD operations on MongoDB, reaching 98.64% statement coverage and 93.75% branch coverage with Jest. Endpoints are documented in Swagger.
- Real-Time Chat:** Real-time multi-room messaging with Node.js, Socket.IO, and WebSockets, JWT authentication via Bearer tokens, and persistence of rooms and messages in MongoDB. Unit and integration tests reached 95% branch coverage with Jest. Endpoints are documented in Swagger.

EDUCATION

Computer Engineering – FIUBA, University of Buenos Aires

Buenos Aires · 2022 – Present · ~75% completed

Relevant coursework: *Algorithms and Programming 1/2/3* · *Operating Systems* · *Computer Organization* · *Computer Networks* · *Databases* · *Software Engineering 1* · *Theory of Algorithms* · *Information Systems Development Management*

Algorithmic practice: 87 algorithms and data structures problems on LeetCode — 76.7th percentile

INTERESTS

AI Agents and Agentic Software Development · Software Architecture and Backend Systems · Financial Systems (ISO 20022)